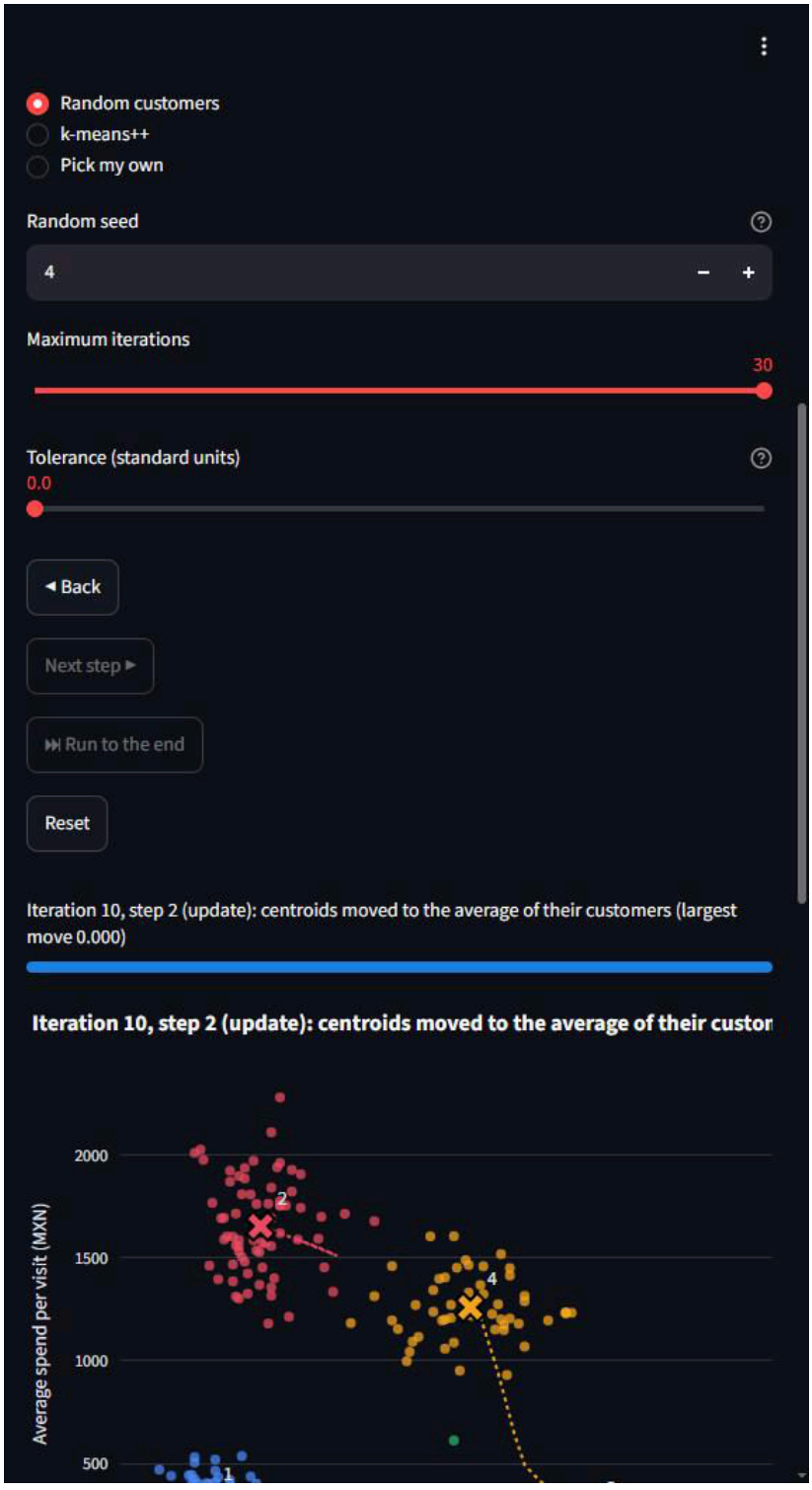
Activity 16 — ShopSmart: Customer Groups with k-means

Labeled screenshots · Session 29 · Diego Kamil Olea Ruiz · Artificial Intelligence

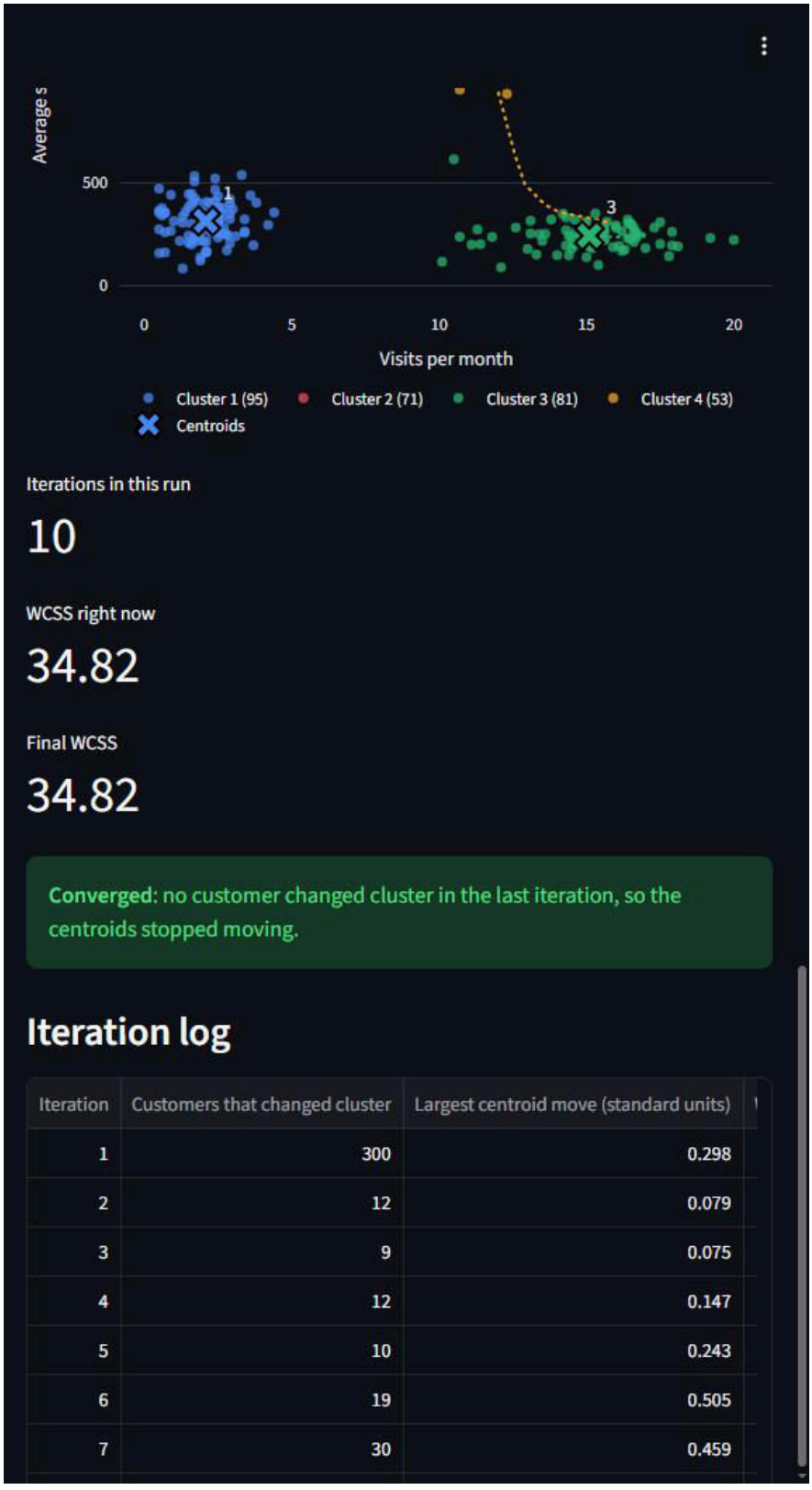
A — Default run, final chart (Task 2, Q2)

Supports question 2. Default settings: k = 4, Random customers, seed 4, max iterations 30, tolerance 0. I ran it to the end and it finished at iteration 10, where the centroids moved 0.000.



B — Default run, stopping message and iteration log (Task 2, Q2 to Q4)

Supports questions 2, 3 and 4. 10 iterations, final WCSS 34.82 and the green Converged message. In the log the largest centroid move is only 0.079 and 0.075 in iterations 2 and 3 but jumps to 0.505 in iteration 6, which is why stopping early is risky.



Iteration	Customers that changed cluster	Largest centroid move (standard units)
1	300	0.298
2	12	0.079
3	9	0.075
4	12	0.147
5	10	0.243
6	19	0.505
7	30	0.459